

Matrix functions and stability

A dimension-independent polynomial norm bound from super-identical pseudospectra

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved**Topic:** Polynomial matrix norms and pseudospectral information

Rating rationale: Existing information controls polynomial norms with a dimension-dependent constant; a uniform comparison would strengthen pseudospectral inference about nonnormal dynamics and polynomial matrix functions.

Resolution: negative, 15 September 2026

Solved negatively by Georg Maierhofer (University of Cambridge). The manuscript is dated 14 September 2026 and passed a fresh independent informal AI-agent audit on 15 September 2026. The exact dimension-dependent constants and their optimal growth remain open.

The [counterexample manuscript](#), **Theorem 1 and equation (15)**, constructs real nonnegative nilpotent weighted shifts of order $N = (m + 1)^2$, for every integer $m \geq 2$ and real $t > 1$. All singular values agree after every complex scalar shift. For the common polynomial $p_m(z) = \sum_{j=1}^m z^{(m+2)^j}$, it proves

$$\frac{\|p_m(X_{m,t})\|_2}{\|p_m(Y_{m,t})\|_2} \geq \frac{\sqrt{m}}{1 + (m-1)/t^2}, \quad p_m(Y_{m,t}) \neq 0.$$

Taking $t = m$ gives a ratio at least $(4/5)\sqrt{m}$, so the original uniform-boundedness target is refuted.

Corollary 3 gives $C_N \geq \sqrt{\lfloor \sqrt{N} \rfloor - 1}$ for $N \geq 9$. It does not determine the sharp C_N or close the gap to the known upper bound.

Evidence level: a complete written argument with reproducible exact and numerical checks, developed and self-reviewed with AI assistance. The [fresh maintainer audit](#) confirms the complete negative resolution and records independent mathematical review and fresh checks. The [submitted informal Codex AI-agent audit](#) is also retained. No external human peer review or Lean verification is asserted. See the [source, checks and provenance](#). The existing difficulty and importance ratings are retained as historical ratings of the original boundedness question.

Context and notation

All matrices are complex and all norms are spectral norms. Write $s_1(M) \geq \dots \geq s_N(M)$ for the singular values of M . Define

$$A \sim_{\text{sip}} B \iff s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N).$$

Define the sharp comparison constant

$$C_N = \sup \left\{ \frac{\|p(A)\|_2}{\|p(B)\|_2} : A, B \in \mathbb{C}^{N \times N}, A \sim_{\text{sip}} B, p \in \mathbb{C}[z], p(B) \neq 0 \right\}.$$

As recorded in [SP-15](#), super-identical pseudospectral data also imply ordinary similarity, so $p(A) = 0$ if and only if $p(B) = 0$.

Problem statement

Determine whether $\sup_{N \geq 1} C_N < \infty$. Equivalently, does there exist an absolute constant C such that

$$\|p(A)\|_2 \leq C \|p(B)\|_2$$

for every dimension, every super-identical-pseudospectral pair and every polynomial? Determining the sharp C_N is the closely related quantitative question, retained here rather than split into another entry.

References

- Fortier Bourque and Ransford, *Super-identical pseudospectra*, p. 513, immediately after Theorem 1.3, explicitly ask both whether their $\sqrt{N}$ bound is optimal and whether a dimension-independent bound exists. The sharp-constant notation above is editorial; the boundedness question is source-stated.
- Thomas Ransford and Nathan Walsh, *A four-mean theorem and its application to pseudospectra*, [arXiv:2109.14472v2](#), 9 July 2022, Theorem 1.3, prove

$$\|p(A)\|_2 < \sqrt{N-2} \|p(B)\|_2 \quad (N \geq 4)$$

unless both sides vanish. **Proposition 5.1** and its following argument show sharpness for $N = 4$: $C_4 = \sqrt{2}$. **Theorem 1.4** concerns unbounded condition numbers of similarity transforms in fixed dimension, a different quantity.

Thus $C_N \leq \sqrt{N-2}$ for $N \geq 4$. The small-dimensional unitary/transpose classification gives $C_1 = C_2 = C_3 = 1$.

Original scope and literature check

The following is the repository's pre-claim literature record, retained unchanged.

The unknown absolute constant, not the superseded $\sqrt{N}$ sharpness guess, is the admission target. This comparison does not follow just from ordinary similarity, since a badly conditioned similarity can change norms substantially. Searches through 2026-09-11 for super-identical polynomial norm bounds, dimension-independent bounds, sharp constants and successors to the four-mean paper found no full solution. The current arXiv version history still ends at v2, 9 July 2022. This is an older-source question supported by bounded searches; no 2026 explicit open-status reaffirmation was located.